

ROWAN ROLARK

Northwestern University, Department of Mechanical Engineering
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EDUCATION

Northwestern University Ph.D. in Mechanical Engineering Specialization: Directed energy deposition, design for additive manufacturing, functionally graded materials	Expected: 06/2027
University of Hawai'i at Mānoa B.S. in Mechanical Engineering Pi Tau Sigma Honor Society GPA: 3.70	05/2022

RESEARCH EXPERIENCE

Graduate Researcher <i>Advanced Manufacturing Processes Laboratory, Northwestern University</i> · PI: Dr. Jian Cao · Developing advanced metal additive manufacturing processes for powder-blown laser-based directed energy deposition · Identifying multi-physic relationships between fundamental process parameters, material properties, and part performance · Implementing a control framework for precise material deposition to fabricate multi-material parts	05/2022 - Present <i>Evanston, IL</i>
Materials Engineering Research Intern <i>Lawrence Livermore National Laboratory</i> · PI: Dr. Maxim Shusteff, Mentor: Michael Triplett · Prototyped a dual-use chamber for printing and cell growth for the development of bioprinted lung models as a platform for studying SARS-CoV-2 infection · Designed a process workflow that integrated computed axial lithography with projection micro-stereolithography to achieve high-resolution, multiscale prints	06/2022 - 08/2022 <i>Livermore, CA</i>
Research Associate <i>Ray Research Group, University of Hawai'i at Mānoa</i> · PI: Dr. Tyler Ray · Led a research project fabricating 3D-printed microfluidic devices for smart wearables that collect and analyze sweat to diagnose diseases · Operated and maintained first-class additive manufacturing platforms (including digital-light processing, stereolithography, and liquid-crystal display 3D printing technologies) · Utilized high-precision metallurgical microscopes for imaging and the characterization of 3D-printed samples	07/2020 - 08/2022 <i>Honolulu, HI</i>
Undergraduate Research Fellow <i>NASA Hawai'i Space Grant Consortium</i> · Co-authored a proposal to the Hawai'i Space Grant Consortium, supported by NASA, to pursue the design and manufacturing of customized life-detection instruments in a simulated Martian rover · Design lead engineer and systems integrator for the rover's science payload involving the collection, preparation, and testing of mineralogical samples for signs of life · Researched and developed robotic methods for soil chemical assays conducted on-site	08/2021 - 07/2022 <i>Honolulu, HI</i>

RESEARCH GRANTS

Design and Development of a Low Cost Life-Detection Payload for the Investigation of a Simulated Martian Environment (\$12,000)

01/2022 - 06/2022

NASA Hawai'i Space Grant Consortium

Honolulu, HI

- PI: Dr. Frances Zhu
- Co-authored a research proposal focused on integrating and implementing life detection instruments (Raman spectroscopy and microfluidic devices) on space exploration rovers. Contributed toward idea conception, guided the project goals, and proposal writing.

PROJECTS

Predictive Maintenance Program

01/2022 - 05/2022

Pearl Harbor Naval Shipyard & Intermediate Maintenance Facility

Honolulu, HI

- Established an innovative predictive maintenance program to monitor and analyze the health of shipyard equipment to prevent machine failures
- Conducted market research on solutions for installing a sensory suite (measuring usage, temperature, vibration, and sound) and analyzing condition monitoring data

FELLOWSHIPS & AWARDS

2026	Leadership Service Award , NSF HAMMER Engineering Research Center
2025	Outstanding Paper in Mfg Systems , North American Manufacturing Conference 53
2025	Leadership Service Award , NSF HAMMER Engineering Research Center
2024	The Graduate School Conference Travel Grant , Northwestern University
2024	Predictive Science and Engineering Design Certificate , Northwestern University
2024	Leadership Service Award , NSF HAMMER Engineering Research Center
2023-2024	Predictive Science and Engineering Design Fellowship , Northwestern University
2022-2023	Walter P. Murphy Fellowship , Northwestern University
2021-2022	NASA Hawai'i Space Grant Consortium Fellowship , University of Hawai'i at Mānoa

COMMUNITY ENGAGEMENT

Volunteer Coordinator

10/2025 - Present

Indigenous Grad Student Collective, Center for Native American and Indigenous Research Evanston, IL

- Spearheaded a project-based learning initiative to build a traditional Native American lodge as a space for teaching, learning, and building community
- Coordinated with the McHenry County Conservation District to organize 25+ volunteers to harvest wood and construct the traditional lodge

President

05/2024 - 09/2025

Student Leadership Council, NSF HAMMER Engineering Research Center

Columbus, OH

- Established infrastructure for new student organization spanning across 5 universities and facilitated communication between students, center administration, and industry partners
- Created alumni program that recognized cohorts of graduating students who were actively engaged in center activities, promoting student involvement and built the foundation for the center's alumni network
- Collaborated between other NSF engineering research centers to build a best practices manual on how to lead successful interdisciplinary engineering research centers

- Organized annual outreach events such as Women in Manufacturing Day, where HAMMER students engaged with 250+ female high school students and exposed them to advanced manufacturing and research opportunities

President

01/2024 - 12/2025

Mechanical Engineering Graduate Student Society, Northwestern University

Evanston, IL

- Managed department events and initiatives by communicating with department leadership and facilitating efforts for student community building

Activity Coordinator

01/2023 - 03/2023

Unseen Identities in STEM, Northwestern University

Evanston, IL

- Organized events centered on uplifting underrepresented minority groups in STEM, such as discussion panels featuring industry professionals and senior graduate students

Diversity, Equity, and Inclusion Seminars

12/2016 - 07/2020

Teach for America

Honolulu, HI

- Lectured and informed 30+ educators on how to be inclusive and how to create safe spaces and address conflict in a classroom setting to support underrepresented students
- Held multiple interactive one-hour seminars at Chaminade University of Honolulu and a local middle school
- Hosted a supplemental question-and-answer session on how to curate a safe environment for underrepresented students

PROFESSIONAL DEVELOPMENT

Research Experience Mentor

06/2024 - 07/2024

Northwestern University

Evanston, IL

- Mentored Eve Maramba, an undergraduate students from the University of Florida. I helped her develop an individual development plan and coached her on how to present research. She was awarded People's Choice Winner of NSF HAMMER-ERC's 2024 Perfect Pitch Competition
- Project: LabVIEW processing of multiple in-situ data streams for directed energy deposition

Equity in STEM Journal Club

04/2024

Searle Center for Advancing Teaching and Learning, Northwestern University

Evanston, IL

- Participated in discussions about equity in STEM for underrepresented groups

Creating Inclusive STEM

09/2023 - 12/2023

Searle Center for Advancing Teaching and Learning, Northwestern University

Evanston, IL

- Organized events centered on uplifting underrepresented minority groups in STEM, such as discussion panels featuring industry professionals and senior graduate students

TECHNICAL SKILLS

Programming Languages

MATLAB, Python, R, C++, G-Code, Java

CAD Modeling

AutoCAD, SolidWorks, Fusion360, Google Sketchup, Meshmixer

Software & Tools

ImageJ, nTopology, ABAQUS, ParaView, Altair

PRESENTATIONS

F. Rolark, R. Zha, K.H. Ryou, J. Jeong, J. Cao, "Developing Powder Flow Control for Precise Functionally Graded Materials via Laser Directed Energy Deposition," *Solid Freeform Fabrication Symposium 2024*, Austin, TX, USA, 2024

F. Rolark, T.R. Ray, "High precision 3D-printed molds for soft lithography of epidermal microfluidic devices," *University of Hawai'i at Mānoa Summer Undergraduate Research Experience Symposium*, Honolulu, HI, USA, 2021

PUBLICATIONS

S. Baruah, J. Dvorak, K. Sargent, R. Patterson, C. Campbell, N. Hall, D. Goodspeed, X. Li, **R. Rolark**, B. Thurston, R. Tomazin, B. Jared, T. Schmitz, T. Sun, A. Bandar, J. Cao, J. Lewandowski, G. Daehn, "Process-level benchmarking of hybrid additive-subtractive manufacturing routes for aerospace structural components," *Journal of Manufacturing Systems*, vol. 86, pp. 648–658, Jun. 2026, doi: 10.1016/j.jmsy.2026.04.007.

T. Wu, R. Zhou, **R. Rolark**, R. Zha, J. Cao, and D. C. Dunand, "Rapid characterization of high-temperature oxidation via X-ray diffraction in laser-deposited superalloys," *Materials Characterization*, vol. 234, p. 116232, Apr. 2026, doi: 10.1016/j.matchar.2026.116232.

K. Ryou, J. Y. Choi, **R. Rolark**, D. C. Dunand, and J. Cao, "Restricted variant selection in martensite transformation of maraging steel during additive manufacturing," *Scripta Materialia*, vol. 270, p. 116961, Jan. 2026, doi: 10.1016/j.scriptamat.2025.116961.

T. Wu, K. Ryou, R. Zha, **R. Rolark**, J. Cao, and D. C. Dunand, "Creep of direct-energy-deposited Inconel 625 with evolving microstructure," *Additive Manufacturing*, vol. 111, p. 104979, Aug. 2025, doi: 10.1016/j.addma.2025.104979.

J. Y. Choi, S. M. Estalaki, D. Quispe, R. Zha, **R. Rolark**, M. Mozaffar, J. Cao, "Transfer learning enabled geometry, process, and material agnostic RGNN for temperature prediction in directed energy deposition," *Additive Manufacturing*, vol. 109, p. 104876, Jul. 2025, doi: 10.1016/j.addma.2025.104876.

Y.-P. Chen, V. Karkaria, Y.-K. Tsai, **F. Rolark**, D. Quispe, R. X. Gao, J. Cao, W. Chen, "Real-time Decision-Making for Digital Twin in Additive Manufacturing with Model Predictive Control using Time-Series Deep Neural Networks," *Journal of Manufacturing Systems*, vol. 80, pp. 412–424, Jun. 2025, doi: 10.1016/j.jmsy.2025.03.009.